VIDYA ACADEMY OF SCIENCE AND TECHNOLOGY

PCCSL507 – NETWORKS LAB

EXPERIMENT 14

Implement OSPF Routing

1 Aim

To configure OSPF (Open Shortest Path First) routing on the routers and verify connectivity using the routing table and ping commands.

2 Requirements

- Cisco Packet Tracer
- Four Cisco 2811 routers
- One PC
- Copper straight-through cable
- Serial DCE/DTE cables

3 Theory

OSPF (Open Shortest Path First) is a dynamic link-state routing protocol used by routers to exchange information about the network topology and learn routes to remote networks.

OSPF uses cost as its routing metric and calculates the shortest path to a destination. It uses Link-State Advertisements (LSAs) to exchange link-state information between neighboring routers.

OSPF divides a network into areas. Area 0 is the backbone area. In this experiment, all networks are configured in Area 0.

OSPF-learned routes are identified by the letter `O` in the routing table. The administrative distance of OSPF is 110.

4 Network Topology

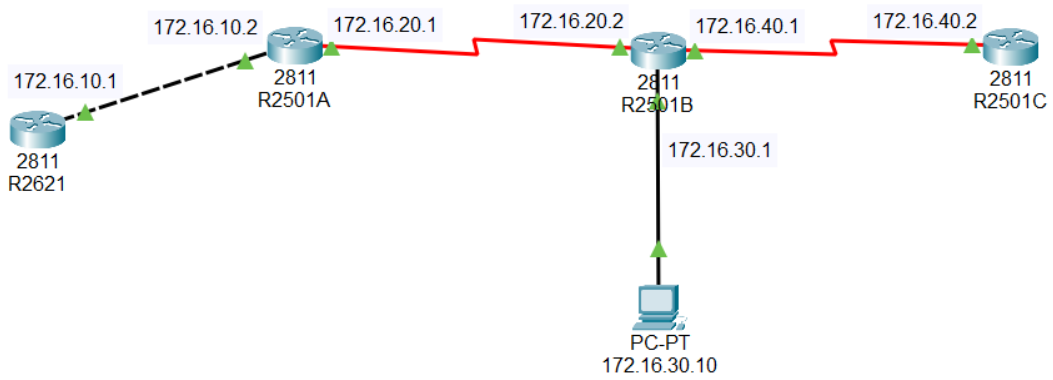


Figure 1: OSPF Network Topology

5 IP Address Allocation

Device	Interface	IP Address	Subnet Mask
R2621	FastEthernet0/0	172.16.10.1	255.255.255.0
R2501A	FastEthernet0/0	172.16.10.2	255.255.255.0
R2501A	Serial0/0/0	172.16.20.1	255.255.255.0
R2501B	Serial0/0/0	172.16.20.2	255.255.255.0
R2501B	FastEthernet0/0	172.16.30.1	255.255.255.0
R2501B	Serial0/0/1	172.16.40.1	255.255.255.0
R2501C	Serial0/0/0	172.16.40.2	255.255.255.0
PC	FastEthernet0	172.16.30.10	255.255.255.0

Table 1: IP Address Allocation

5.1 Network Addresses

Network	Network Address
R2621 – R2501A	172.16.10.0/24
R2501A – R2501B	172.16.20.0/24
R2501B – PC	172.16.30.0/24
R2501B – R2501C	172.16.40.0/24

Table 2: Network Addresses

6 Connectivity

Device	Interface	Connected Device	Interface
R2621	FastEthernet0/0	R2501A	FastEthernet0/0
R2501A	Serial0/0/0 (DCE)	R2501B	Serial0/0/0 (DTE)
R2501B	Serial0/0/1 (DCE)	R2501C	Serial0/0/0 (DTE)
R2501B	FastEthernet0/0	PC	FastEthernet0

Table 3: Network Connectivity

7 Procedure

1. Connect the routers and PC according to the given topology.
2. Assign IP addresses to all router interfaces.
3. Configure the DCE interfaces with the required clock rate.
4. Enable all required interfaces using `no shutdown`.
5. Configure the IP address, subnet mask, and default gateway on the PC.
6. Verify the interface status using `show ip interface brief`.
7. Configure OSPF on all routers.
8. Verify OSPF neighbor relationships.
9. Display the OSPF routing table.
10. Verify the OSPF database.
11. Verify connectivity using the `ping` command.

8 Router IP Configuration

8.1 Router R2621

```
enable
configure terminal

interface fastethernet0/0
ip address 172.16.10.1 255.255.255.0
no shutdown
exit

end
```

8.2 Router R2501A

```
enable
configure terminal

interface fastethernet0/0
ip address 172.16.10.2 255.255.255.0
no shutdown
exit

interface serial0/0/0
ip address 172.16.20.1 255.255.255.0
clock rate 64000
no shutdown
exit

end
```

8.3 Router R2501B

```
enable
configure terminal

interface serial0/0/0
ip address 172.16.20.2 255.255.255.0
no shutdown
exit

interface fastethernet0/0
ip address 172.16.30.1 255.255.255.0
no shutdown
exit

interface serial0/0/1
ip address 172.16.40.1 255.255.255.0
clock rate 64000
no shutdown
exit

end
```

8.4 Router R2501C

```
enable
configure terminal

interface serial0/0/0
ip address 172.16.40.2 255.255.255.0
no shutdown
```

```
exit

end
```

9 PC Configuration

Configure the PC using the following values:

```
IP Address      : 172.16.30.10
Subnet Mask     : 255.255.255.0
Default Gateway : 172.16.30.1
```

10 OSPF Configuration

10.1 Router R2621

```
enable
configure terminal

router ospf 1
network 172.16.10.0 0.0.0.255 area 0

end
```

10.2 Router R2501A

```
enable
configure terminal

router ospf 1
network 172.16.10.0 0.0.0.255 area 0
network 172.16.20.0 0.0.0.255 area 0

end
```

10.3 Router R2501B

```
enable
configure terminal

router ospf 1
network 172.16.20.0 0.0.0.255 area 0
network 172.16.30.0 0.0.0.255 area 0
network 172.16.40.0 0.0.0.255 area 0

end
```

10.4 Router R2501C

```
enable
configure terminal

router ospf 1
network 172.16.40.0 0.0.0.255 area 0

end
```

The OSPF network statements and Area 0 configuration follow the uploaded Experiment 14 procedure. :contentReference[oaicite:1]index=1

11 Verification

11.1 Interface Verification

Verify the interface status on each router:

```
show ip interface brief
```

All connected interfaces should be in the `up/up` state.

11.2 OSPF Neighbor Verification

Use the following command:

```
show ip ospf neighbor
```

A neighbor state of `FULL` indicates that the OSPF adjacency has been successfully established.

11.3 Routing Table Verification

Display the routing table using:

```
show ip route
```

OSPF-learned routes are identified by `O`.
For example:

```
C 172.16.10.0/24
O 172.16.20.0/24
O 172.16.30.0/24
O 172.16.40.0/24
```

Here,

- `C` = Connected Route
- `O` = OSPF Learned Route
- `110` = Administrative Distance of OSPF

The experiment sheet uses the same OSPF route codes and administrative distance in its sample routing table. :contentReference[oaicite:2]index=2

11.4 OSPF Database Verification

Use:

```
show ip ospf database
```

This command displays the Link-State Advertisements (LSAs) exchanged among the OSPF routers. :contentReference[oaicite:3]index=3

11.5 Connectivity Verification

From R2621:

```
ping 172.16.30.1
ping 172.16.40.2
```

From R2501C:

```
ping 172.16.10.1
```

From PC:

```
ping 172.16.30.1
ping 172.16.10.1
ping 172.16.40.2
```

Successful replies confirm connectivity between the different networks. The uploaded experiment also uses the router-to-router ping tests shown above. :contentReference[oaicite:4]index=4

12 Result

OSPF routing was successfully configured on all routers. The routers established OSPF neighbor relationships and dynamically learned the remote networks. The routing table displayed OSPF-learned routes, and connectivity between the networks was successfully verified using the ping command.